

Question Paper

Exam Date & Time: 21-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Artificial Intelligence [CSE_3128]

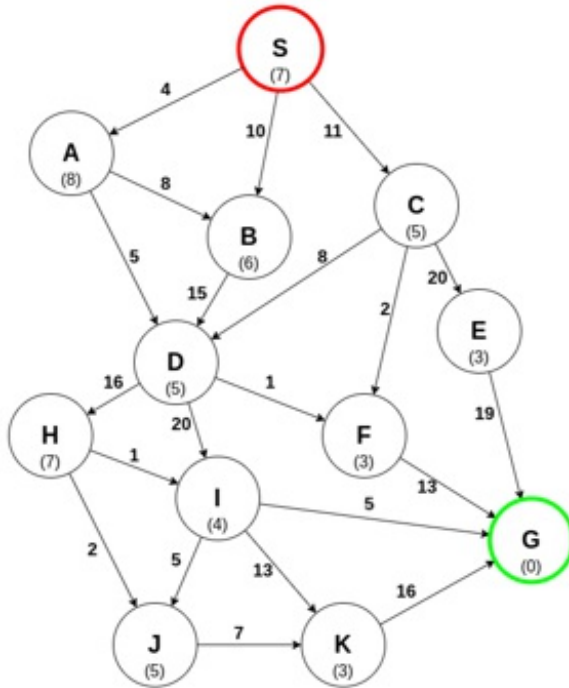
Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

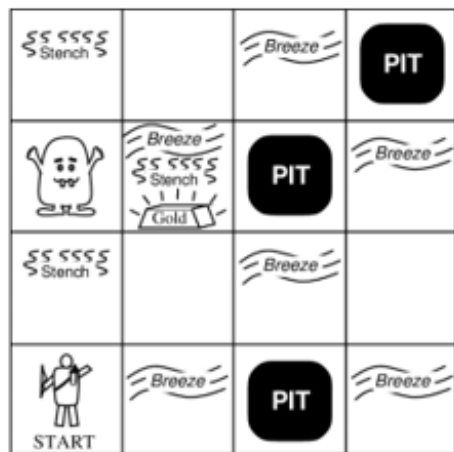
- 1) Compare and analyze the performance of goal-based and utility-based agents in dynamic environments. Discuss how their decision-making processes differ in their applications with the help of examples. (5)
- A)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Compare and contrast between formal task and expert task in the domain of artificial intelligence. Apply these themes and present suitable examples. (5)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Analyze the states PAGE/ PEAS concept to the autonomous vacuum cleaner. (5)
- A)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Compare among fully observable and partially observable environments in AI with example. (5)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Analyse the pros and cons of DFS and BFS for water-jug problem solving. (5)
- A)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Apply A* algorithm for your AI agent to perform the search on the graph given below: (5)



Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Do the knowledge representation for the Wumpus World and analyze how W13 can be proved. (5)

A)



Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Convert the following into clausal form (Analyze all the steps): (5)
 $\exists u \exists v \forall w (\exists z P(f(u, w), v, z) \rightarrow (\exists x Q(u, x) \& \exists y R(v, y)))$

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Why is it preferred to apply alpha-beta pruning over min-max strategy? Also, provide an example with respect to alpha-beta pruning strategy. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Apply Predicate logic to represent the following sentences: (5)
- I) Every man respects his parent.
 - II) There exists at least one student who does not like both Mathematics and Biology.
 - III) Every Indian whose yearly income exceeds ₹60,000 is required to pay income tax.
 - IV) There exists some girl who participates in cricket.
 - V) Rose has thorns.

-----End-----

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 CSE_3128:Artificial Intelligence

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	15
4	4	CO 4	10
5	5	CO 5	5
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 CSE_3128:Artificial Intelligence

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	4	20(40%)
4	Analysis	6	30(60%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)